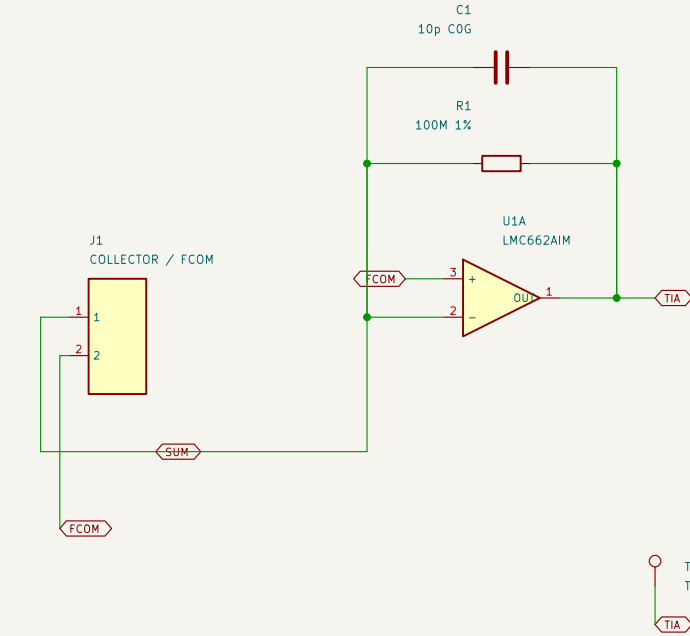


	1	2	3	4	5	6	7	8
A	Floating biased–collector detector / 12 or 24 V input							
B	Guarded LMC662 current acquisition / ADS1115 J1 SUM → guarded LMC662 → conditioning → ADS1115. 100 Mohm feedback, 10 pF; all local voltages reference FCOM. Preserved 16 mm SUM trace; no copper plane under the input. File: collector12_analog.kicad_sch 9–30 V input / onboard buck / isolated converter J2 regulated 9–30 V DC → fuse → reverse diode → buck. LMR51420XF: 500 kHz forced PWM; 5.08889 V nominal. R05P09D/P provides isolated raw positive/negative rails. File: collector12_power.kicad_sch Filtered local +/–5 V detector rails Raw +/–9 V → preload and RC filters → TPS7A regulators. Quiet +5VA and –5VA reference FCOM; local 3.3 V for ADC. 55 mA per raw rail; verify ripple, headroom and temperature. File: collector12_power_rails.kicad_sch Onboard regulated collector bias / controls and discharge TPS2553 latch–off switch supplies CA02P–5; U23 senses output. Supply–ready AND host release enable the filtered manual setpoint. 0–2.048 V program: nominal 0–200 V raw bias; default clamped zero. File: collector12_bias.kicad_sch Floating ATmega acquisition / bidirectional optical UART ATmega328P at 3.3 V / 8 MHz acquires ADS1115 samples. Two wide DIP optical channels carry 38400 baud UART. HOST_VIO is a 3.3 V input; isolated ICSP requires discharged bias. File: collector12_readout.kicad_sch Collector–bias filtering / bleeders / guarded current return Two 10 kohm / 1 W filters feed FCOM through rated capacitors. Raw and final bleeders remain independent of host software. Collector connects only to SUM; rated discharge check is required. File: collector12_bias_filter.kicad_sch							
C	POWER AND REFERENCE Input: regulated 9–30 V DC, nominal 12 or 24 V. Buck: 5.08889 V nominal; divider resistors 0.1%. Analog: +5VA / –5VA and local 3.3 V. FCOM floats with the collector bias; PGND is chamber ground. ESP32 is powered separately. No gun or filament connection. SIGNAL AND CONNECTORS J1: 1 SUM, 2 FCOM inner guard, J4: chamber / outer shield. J2: 1 VIN, 2 PGND, J3: 1 GND, 2 HOST_VIO input, 3 host TX, 4 host RX, 5 ENABLE, 6 RELEASE, 7 FAULTn, 8 GND. J5: standard 2x3 AVR ICSP; physically disconnect bias first. MECHANICAL AND VALIDATION 220 x 140 mm, two copper layers, four 3.2 mm NPTH holes. Use insulating nylon / PEEK fasteners and standoffs. Native ERC/DRC and pin–map audits accompany the package. These checks do not establish measured analog performance. UNBUILT ENGINEERING PROTOTYPE Validate leakage, noise, stability, settling and calibration. Inspect construction and clean the high–impedance input. No measured sensitivity or microscope resolution is claimed. Basic 250 VAC RMS working converter isolation. 5 mm copper corridor; not a certified safety barrier. Verify raw HV and FCOM discharged before handling.							
D								
E	Unbuilt engineering prototype / see ../README.md Engineering prototype – see design notes and verified interfaces Integrated detector prototypes Sheet: / File: collector12.kicad_sch Title: Floating biased–collector detector / 12 or 24 V input Size: A3Date: 2026–09–13Rev: B0 – UNBUILT KiCad E.D.A. 10.0.4Id: 1/7							
F	1	2	3	4	5	6	7	8

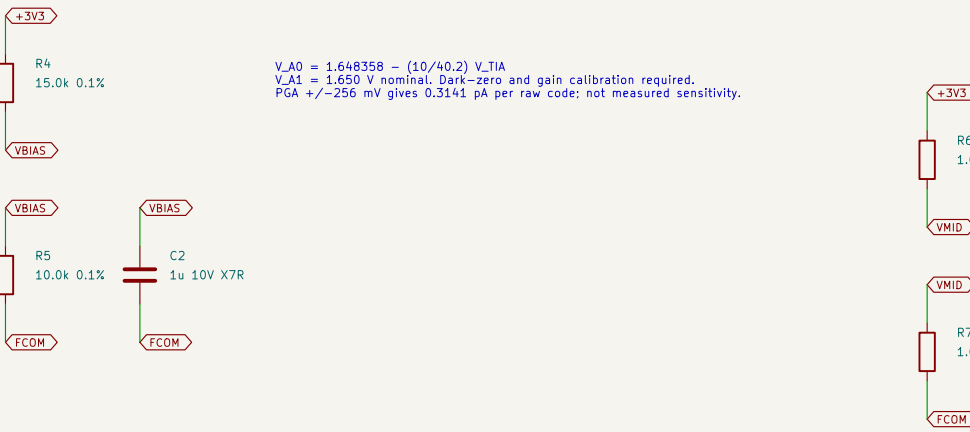
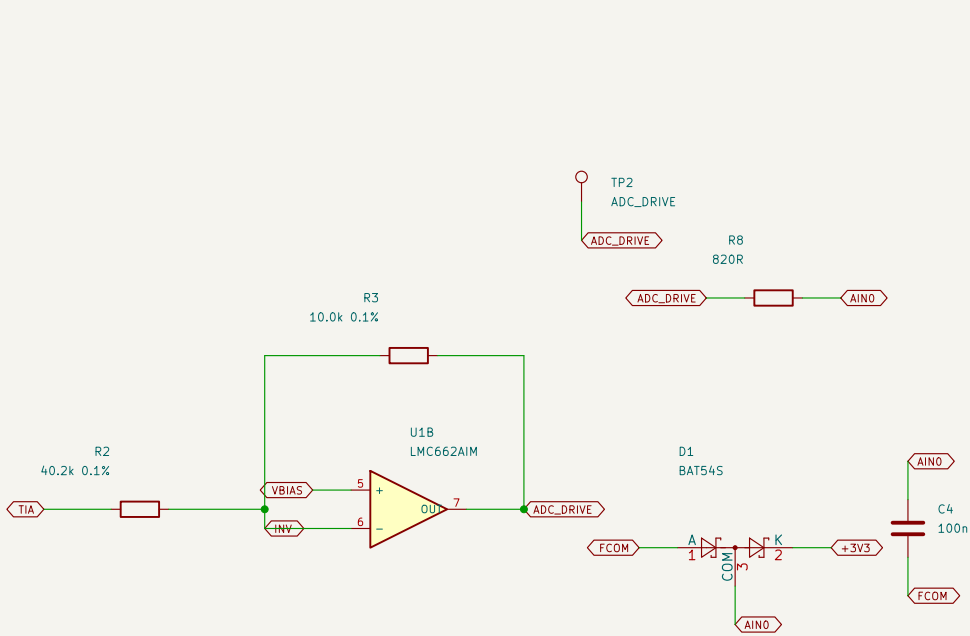
01 GUARDED TRANSIMPEDANCE

02 ADC CONDITIONING

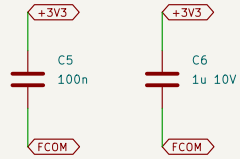
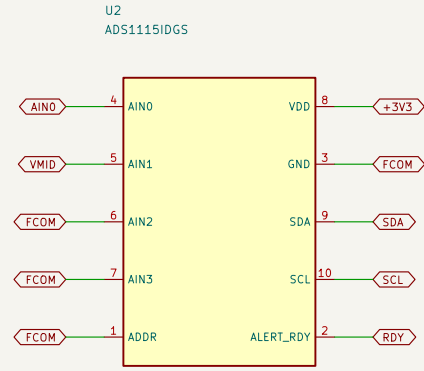
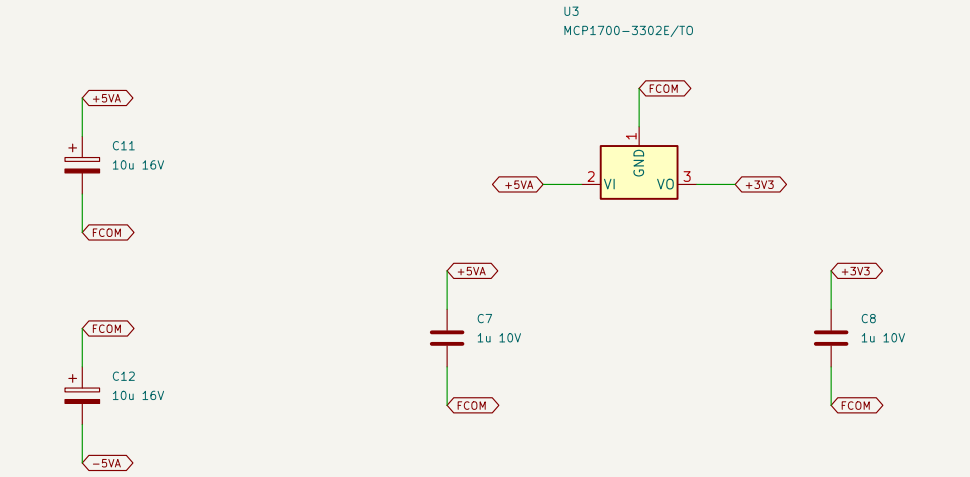
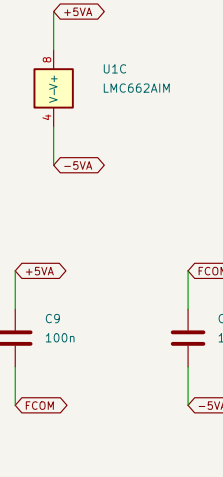
03 DIFFERENTIAL ADC



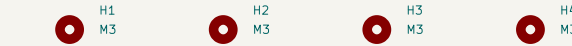
All voltages on this sheet reference FCOM.
SUM has no clamp or probe point. R1 || C1: 1 ms nominal.
I_{abs} electron deposition gives V_{TIA} = +I_{abs} x 100 Mohm.
Do not connect SUM directly to the bias supply.



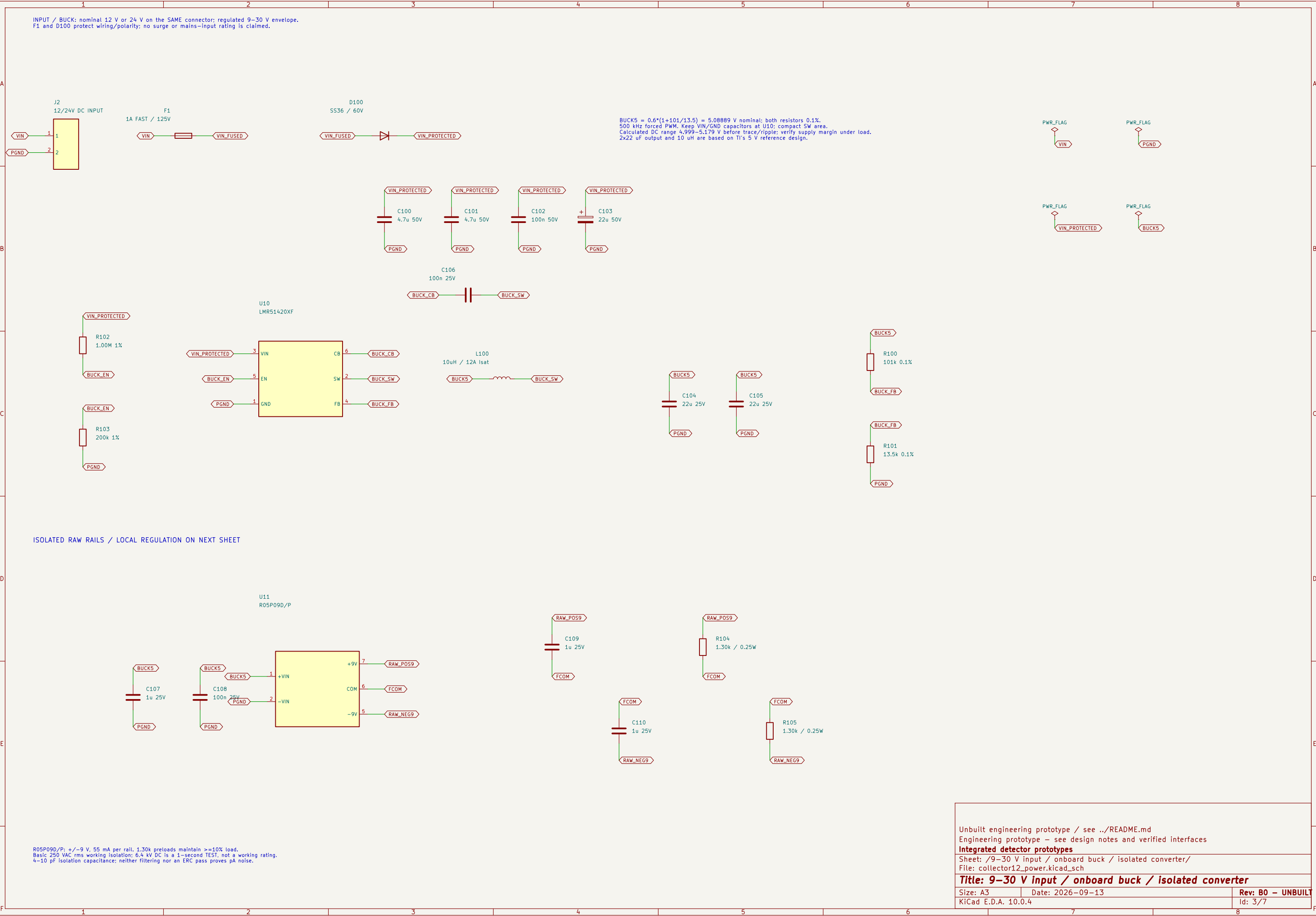
04 LOCAL BYPASS / QUIET ADC REFERENCE

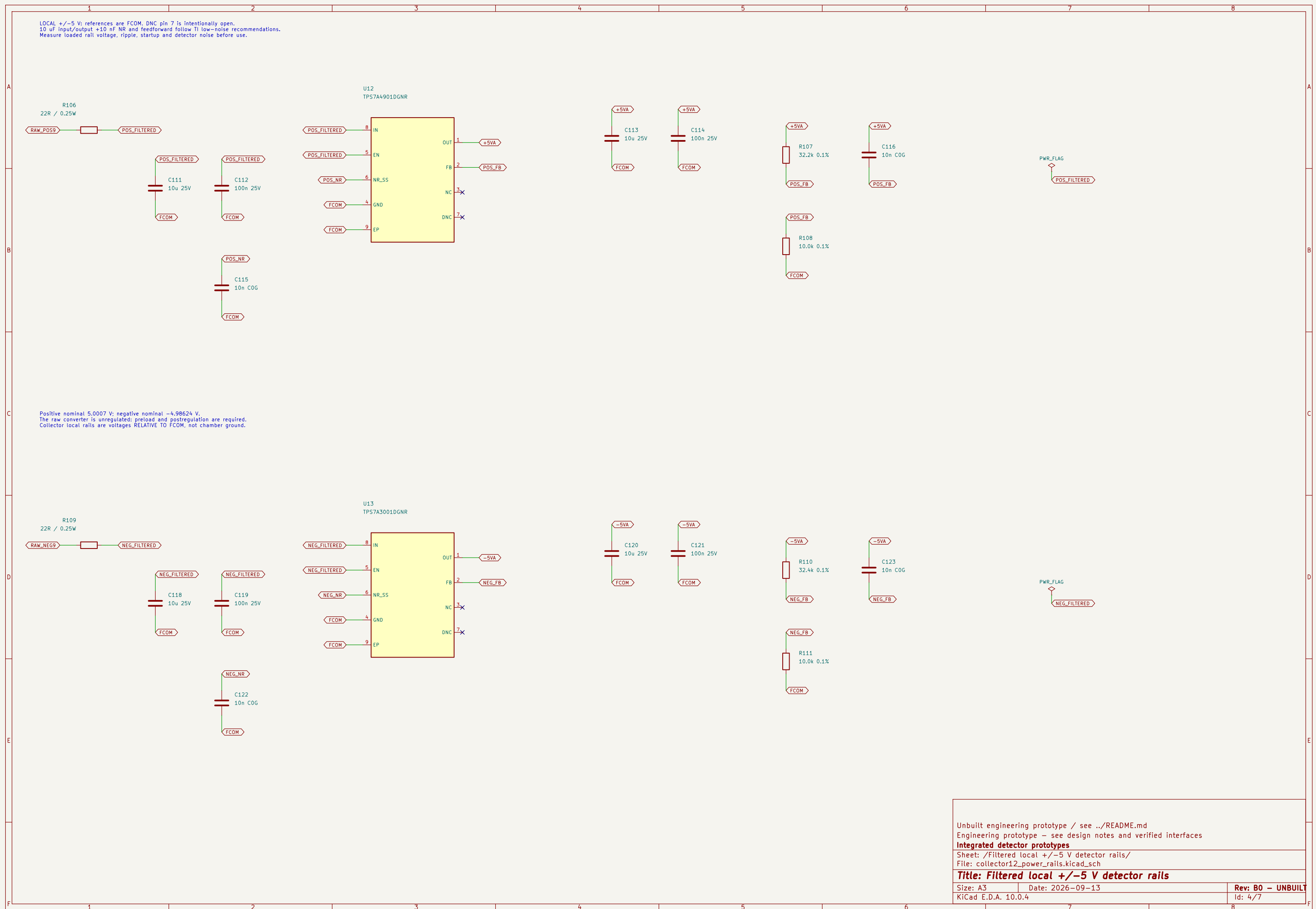


UNBUILT PROTOTYPE
No measured leakage, noise, stability, settling, or imaging resolution.
LMC662 and feedback geometry preserved from the original detector.



Unbuilt engineering prototype / see ../README.md		
Engineering prototype – see design notes and verified interfaces		
Integrated detector prototypes		
Sheet: /Guarded LMC662 current acquisition / ADS1115/		
File: collector12_analog.kicad_sch		
Title: Guarded LMC662 current acquisition / ADS1115		
Size: A3	Date: 2026-09-13	Rev: B0 – UNBUILT
KiCad E.D.A. 10.0.4		Id: 2/7





1 DEFAULT OFF
HV_ENABLE low: U22 removes module 5 V. Fault latches off.
Either control low: 2N7002 clamps filtered command to 0 V.
Q203/Q204 require PGM_RELEASE AND actual supply-ready.

2 SEQUENCED ENABLE
Apply module power with command held zero. U23 holds reset 180–420 ms after actual module supply rises.
Raise PGM_RELEASE: 10k / 47uF gives 0.47–0.588 s nominal ramp including pot resistance.
Accept C210 only with effective C >=22uF at 2,048 V / temperature; measure settling.
Disable by releasing command first, then removing module power.

A

B

C

D

E

F

A

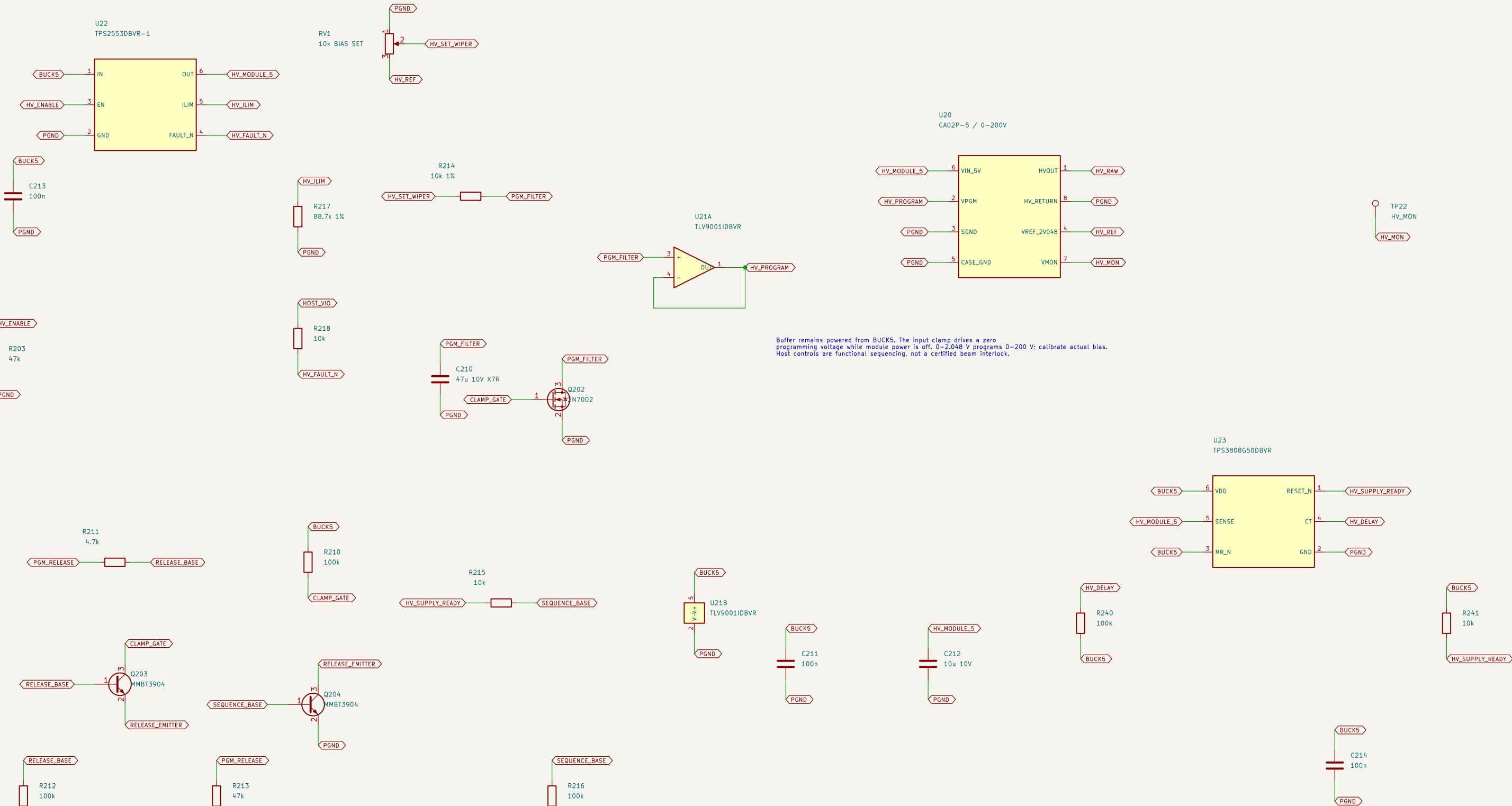
B

C

D

E

F



Buffer remains powered from BUCK5. The input clamp drives a zero programming voltage while module power is off. 0–2,048 V programs 0–200 V; calibrate actual bias. Host controls are functional sequencing, not a certified beam interlock.

CA02P-5 is ONBOARD, not an external supply. Nonisolated grounds 3/8; case 5 grounded.
Hold VPGM=0 for >=100 ms after module power, then ramp. Inhibit/removal of input and independent bleeders must be implemented on this sheet. Monitor raw HV AND FCOM.
Nominal 200 V module becomes about 199.6 V after 20k filter/9.98M bleeder; no 300 V claim.

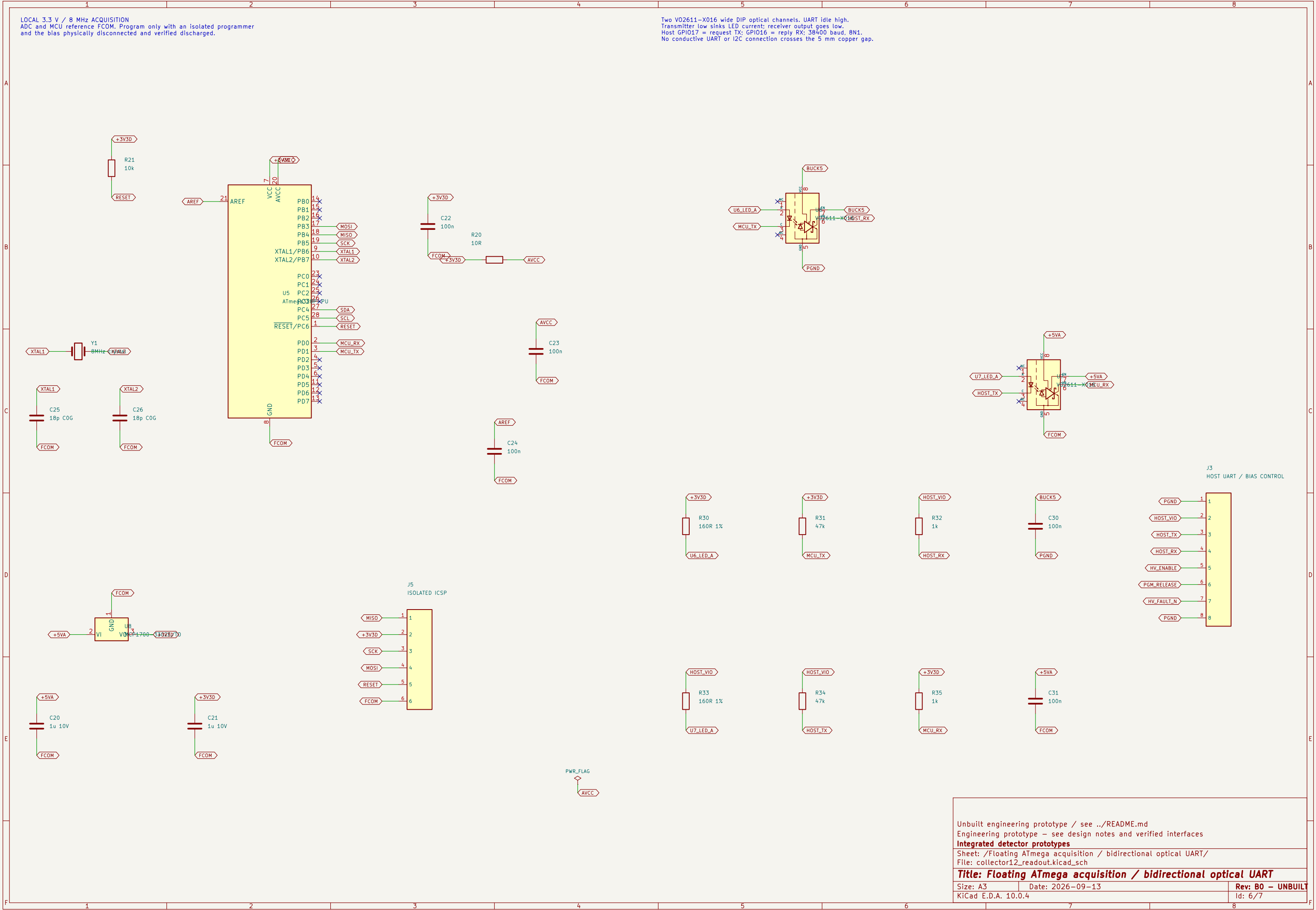
Unbuilt engineering prototype / see ../README.md
Engineering prototype – see design notes and verified interfaces
Integrated detector prototypes
Sheet: /Onboard regulated collector bias / controls and discharge/
File: collector12_bias.kicad_sch

Title: Onboard regulated collector bias / controls and discharge

Size: A3 Date: 2026–09–13 Rev: B0 – UNBUILT
KiCad E.D.A. 10.0.4 Id: 5/7

LOCAL 3.3 V / 8 MHz ACQUISITION
ADC and MCU reference FCOM. Program only with an isolated programmer
and the bias physically disconnected and verified discharged.

Two VO2611–X016 wide DIP optical channels. UART idle high.
Transmitter low sinks LED current; receiver output goes low.
Host GPIO17 = request TX; GPIO16 = reply RX; 38400 baud, 8N1.
No conductive UART or I2C connection crosses the 5 mm copper gap.



RAW HV -> 10k -> HV_FILTER -> 10k -> FCOM
 100nF/400V at HV_FILTER; 1uF/400V from FCOM to grounded PGND.
 FCOM is the amplifier reference. Connect the collector ONLY to SUM.

Nominal FCOM = $200 \times 9.98M / (9.98M + 20k) = 199.6008 \text{ V}$.
 Independent 9.98M bleeder gives 9.98 s nominal capacitor time constant.
 Whole-board discharge depends on module/cable capacitance and must be measured.
 Measure raw HV and FCOM with rated instruments before touching or connecting ICSP.

Unbuilt engineering prototype / see ../README.md		
Engineering prototype – see design notes and verified interfaces		
Integrated detector prototypes		
Sheet: /Collector–bias filtering / bleeders / guarded current return/ File: collector12_bias_filter.kicad_sch		
Title: Collector–bias filtering / bleeders / guarded current return		
Size: A3	Date: 2026–09–13	Rev: B0 – UNBUILT
KiCad E.D.A. 10.0.4		Id: 7/7

RAW HV -> 10k -> HV_FILTER -> 10k -> FCOM
 100nF/400V at HV_FILTER; 1uF/400V from FCOM to grounded PGND.
 FCOM is the amplifier reference. Connect the collector ONLY to SUM.

Nominal FCOM = $200 \times 9.98M / (9.98M + 20k) = 199.6008 \text{ V}$.
 Independent 9.98M bleeder gives 9.98 s nominal capacitor time constant.
 Whole-board discharge depends on module/cable capacitance and must be measured.
 Measure raw HV and FCOM with rated instruments before touching or connecting ICSP.

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Size: A3	Date: 2026–09–13	Rev: B0 – UNBUILT
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Unbuilt engineering prototype / see ../README.md
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Integrated detector prototypes
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Title: Collector–bias filtering / bleeders / guarded current return

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	7	8

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